

# Practical Applications of TmMgGaO<sub>4</sub>

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**Introduction.** TmMgGaO<sub>4</sub> is most relevant as a **specialised cryogenic magnetic material**. Its strongest uses exploit three measurable properties: strong directional magnetic response, sharp field and temperature dependent changes and tunable collective spin excitations.

The most credible first products are therefore **calibration components, threshold sensors, and cryogenic resonant elements**, rather than room-temperature memory or general-purpose electronics.

## 1. Selected Applications

### 1. Calibration standard for millikelvin magnetic instruments

**Where:** Dilution-refrigerator AC susceptometers in quantum-material laboratories and cryogenic-instrument companies.

**What:** Incorrect temperature calibration, magnetic-field strength, crystal alignment, or pickup-coil sensitivity.

**How:** A pre-characterised, oriented crystal is measured against certified field-temperature response curves.

**Advantages:** One compact sample tests several instrument functions at once, whereas a simple paramagnetic standard mainly checks signal amplitude.

**Direction:** Sell matched crystals with reference curves, orientation marks, and an uncertainty range.

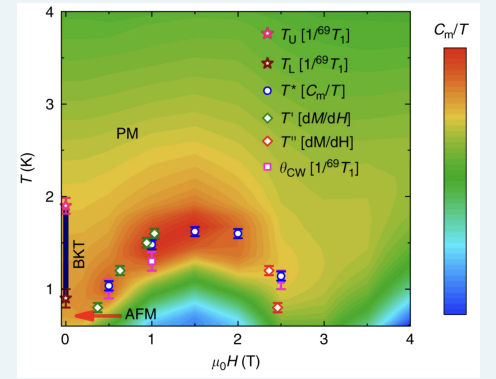


Figure 1: Field-temperature phase diagram of TmMgGaO<sub>4</sub>. Adapted from Hu *et al.*, *Nat. Commun.* **11**, 5631 (2020), CC BY 4.0.

### 2. Field-uniformity and overshoot sensor for superconducting magnets

**Where:** Fusion-magnet test stands, accelerator dipole prototypes, high-field NMR magnets, and pulsed-field laboratories.

**What:** Local field gradients, coil misalignment, uneven energisation, or whether a pulse crossed a preset field threshold.

**How:** Several oriented crystals are placed around the coil. Their susceptibility or resonant response changes when the local field reaches the calibrated magnetic regime.

**Advantages:** It provides a passive, directional, material-based threshold check that can remain inside the cryogenic bore and complement Hall probes, pickup coils, and voltage taps.

**Direction:** Build a three- or four-sensor array and compare its response with a calibrated Hall probe.

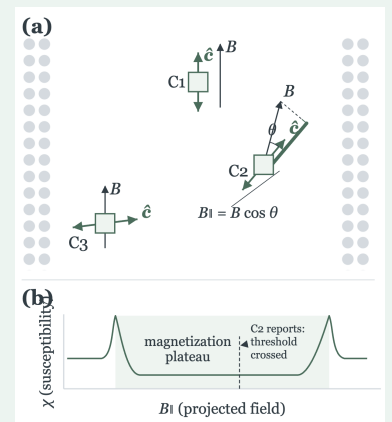


Figure 2: (a) Each crystal responds only to  $B_{\parallel} = B \cos \theta$  along its Ising axis  $\hat{c}$ . (b) Susceptibility peaks bound the magnetization plateau. Figure generated by Claude AI

### 3. Narrow-range thermometer for superconducting quantum hardware

**Where:** Beside superconducting qubit chips, Josephson amplifiers, or transition-edge sensors inside a dilution refrigerator.

**What:** Local heating from microwave-control pulses, poor thermal anchoring, or temperature drift near a chosen subkelvin operating point.

**How:** The crystal is thermally attached to the device and inductively coupled to a microwave resonator. Temperature-dependent susceptibility shifts the resonant frequency.

**Advantages:** Near a magnetic transition it could provide a sharper local response than a broad-range thermometer, while requiring no biased semiconductor junction at the sensing point.

**Direction:** Measure sensitivity, noise, response time, magnetic cross-sensitivity, and self-heating.

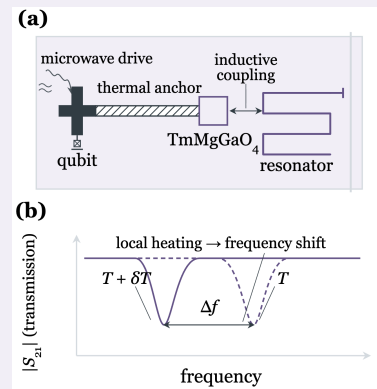


Figure 3: (a) A TmMgGaO<sub>4</sub> crystal is thermally coupled to the target device and inductively coupled to a microwave resonator. (b) Heating changes the magnetic susceptibility. Figure generated by Claude AI.

### 4. Reference crystal for neutron and nonlinear THz instruments

**Where:** Low-energy neutron spectrometers, high-field sample environments, and two-dimensional THz spectroscopy laboratories.

**What:** Energy-resolution errors, field misalignment, polarization errors, pulse-timing errors, or failure to resolve collective magnetic modes.

**How:** The measured spectrum is compared with established field-dependent peaks and collective excitations of the crystal.

**Advantages:** Unlike a single-line calibration sample, it can test field, polarization, energy resolution, and many-body spectral response using one target.

**Direction:** Package oriented crystals with reference spectra for specific field and temperature ranges.

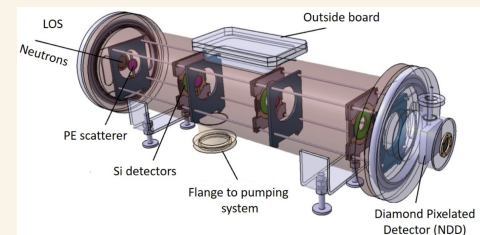


Figure 4: Conceptual design of a high resolution neutron spectrometer. Marek Scholz et al 2019 Nucl. Fusion 59 065001

## 5. Field-tunable cryogenic microwave resonator or filter

**Where:** Cryogenic readout chains for superconducting detectors, qubits, or radio-astronomy receivers.

**What:** Tunes a resonance, creates a selectable stop band, or changes microwave transmission when the magnetic field is adjusted.

**How:** A TmMgGaO<sub>4</sub> crystal is coupled to a coplanar waveguide or cavity. Its field-dependent magnetic excitations alter the resonator response.

**Advantages:** The operating frequency can be changed after fabrication, and the same material may provide nonlinear or threshold behaviour unavailable from a passive dielectric.

**Direction:** Measure insertion loss, tunability, linewidth, power handling, and compatibility with superconducting circuits.

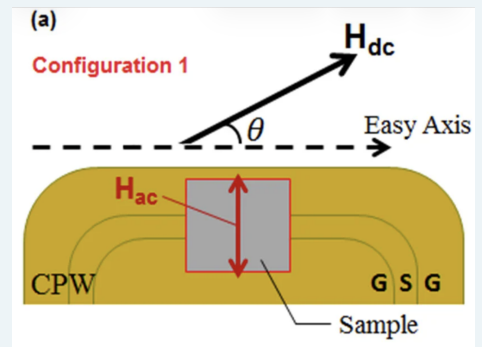


Figure 5: Schematic representation of sample on CPW and orientation of the magnetic field. Daniel J Adams et al J. Phys. D: Appl. Phys. 51 (2018) 055005

## 2. Limitations and Future Directions

### Limitations

#### Interactions must match

Many materials require Heisenberg exchange, spin-orbit coupling, or other complex interactions that cannot yet be implemented directly with neutral-atom quantum processors.

#### The spin representation must match

Each neutral atom represents an effective spin- $\frac{1}{2}$  particle. Materials with multiple active electronic levels or higher-spin behaviour cannot generally be mapped one-to-one onto today's hardware.

#### System size and realism

Real materials also contain defects instead of being idealised finite systems. Impurities and domain boundaries influence their behaviour.

### Outlook

#### More programmable Hamiltonians

Improved laser control, Rydberg dressing and Floquet engineering could realise a wider range of spin interactions.

#### Hybrid analog-digital simulation

Combining native analog evolution with programmable quantum gates would allow additional interaction terms and more complex quantum behaviour to be incorporated, extending the range of accessible materials.

#### Larger quantum twins

Increasing the number of atoms and improving gate fidelity will enable larger, more realistic simulations that include disorder, finite-size effects and crystal imperfections.

**Why TmMgGaO<sub>4</sub>?** TmMgGaO<sub>4</sub> is a rare example of a material that already matches the native transverse-field Ising Hamiltonian of a Rydberg atom array. This enables a genuine one-to-one quantum twin, whereas most materials will require further advances in programmable interactions and quantum hardware.